

THEJUS MAHAJAN PH.D.

Bioinformatics & Data Scientist

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Google Scholar



CAREER INTERESTS

Computational Scientist (PhD) with 5+ years of experience in **large-scale data analysis**, **statistical modeling**, and **pipeline development**. Trained in **R (Bioconductor, Tidymodels)** and **Python (Pandas, NumPy, Scikit-learn)** for high-dimensional data analysis. Built a viral genomics pipeline using **Nextflow (DSL2)**. Currently deepening skills in data pipeline development, **genomics**, AI/machine learning, and NGS analysis through specialized training and an industry internship. Eager to develop expertise in **single-cell multi-omics** and apply computational and data science skills to advance research in **stem cell biology and genomics**.

EXPERIENCE

Internship – Bioinformatics & Data Pipeline Engineering

HealthTwist GmbH

02/2026 - 04/2026

Berlin, Germany

Clinical data research and development under Dr. Andreas Busjahn

- Refactored a monolithic 1,834-line R data pipeline (tidyverse) for Germany's interventional radiology quality registry (DeGIR); 143,000+ patient records from ~300 clinics. Reduced to 1,348 lines (–26.5%) with byte-identical output verified via `identical()`.
- Externalised 257 hard-coded correction rules into 8 configuration CSV files, enabling non-programmer editing of business rules.
- Replaced 357 deprecated function calls across 4 pipeline files, modernising the codebase to current tidyverse standards.
- Created the `degirtools` R package (9 exported functions, roxygen2 docs, devtools::check() passing) by extracting reusable logic from a 767-line helper file.
- Discovered 2 pre-existing bugs through systematic code analysis; documented both and preserved original behaviour for supervisor review – demonstrating production-code discipline.
- Built an interactive R Shiny dashboard (6 modules: overview, interventions, complications, radiation doses, success rates, about) using plotly, DT, and bslib. Deployed to shinyapps.io with GDPR-safe synthetic data.
- Wrote Tidymodels teaching materials (Quarto/GitHub) replacing the legacy caret framework for ML workflows in R.

Further Training – Bioinformatics and Biostatistics

CQ Beratung + Bildung GmbH

08/2025 - 02/2026

Berlin, Germany

Application-related Bioinformatics and Biostatistics

- Programming methodology: Shell (BASH), Python (object-oriented), R.
- Bioinformatics databases (NCBI, Biopython, SQL) and sequence analysis.
- NGS analysis pipelines (Nextflow, Galaxy); structural bioinformatics.
- Clinical biostatistics (R/Bioconductor, ANOVA, PCA, HCA, survival analysis); GDPR/BDSG compliance.
- Capstone: Built a reproducible Nextflow (DSL2) pipeline for Hepatitis Delta Virus sequence analysis – automated NCBI download, MAFFT alignment, and trimAl trimming with Conda-managed dependencies (🔗 GitHub).

Guest Scientist

Helmholtz-Zentrum Hereon, Ökosystemmodellierung

05/2025 - 10/2025

Geesthacht, Germany

Computational Biologist & Systems Modeler

- Spearheaded research modeling evolutionary processes in marine ecosystems.

STRENGTHS



Data Pipeline Engineering

Refactored a production data pipeline (1,834 lines, 143K records, ~300 clinics) with 26.5% code reduction. Built `degirtools` R package. Byte-level verification after every change via `identical()`.



Complex Problem-Solving

Resolved a critical data-loss error by creating a Fortran simulation to correct for a misplaced particle detector, recalculating results using multi-sensor data.



Bioinformatics & Biostatistics

Bridged physical science expertise with modern biological workflows. Skilled in NGS data analysis, sequence alignment, and multivariate statistical modeling using R (Bioconductor) and Python.

TECHNICAL SKILLS

Programming

Python (Pandas, NumPy, Scikit-learn, Biopython), R (Bioconductor, Tidymodels, ggplot2), SQL, BASH

Bioinformatics

Nextflow (DSL2), NGS Analysis, Multiple Sequence Alignment (MAFFT), Sequence Trimming (trimAl), NCBI / Entrez

Biostatistics & ML

Multivariate Analysis, PCA, ANOVA, Dimensionality Reduction, Clustering, Scikit-learn, Tidymodels

Data Engineering

Large dataset processing (HDF5, NetCDF), Pipeline development, Data visualization, R Shiny, devtools/roxygen2

Tools

Linux/Unix, Git, HPC clusters, Conda

TRAINING / COURSES

IBM: Machine Learning with Python; Introduction to AI

Coursera (in progress)

High Performance Computing (HPC) Training

Jülich Supercomputing Center (JSC),
Forschungszentrum Jülich

MITx: 7.00x Introduction to Biology - The Secret of Life

Comprehensive coursework on genetics, biochemistry,

- Applied advanced computational techniques to simulate long-term environmental impacts.
- Finalised the computational modelling framework and prepared the simulation data for peer-reviewed publication.

Job Search & German Language Learning

Hamburg, Germany

02/2025 - 04/2025 Hamburg, Germany

Active job search and German language acquisition (Goethe B1 certification).

Post-doctoral Scientist

University of Hamburg, Institute of Marine Ecosystem and Fisheries Sciences

08/2021 - 01/2025 Hamburg, Germany

Marine Ecosystem Modelling

- Developed the novel “Cyanobacteria Life Cycle” (CLC) model within the ERGOM framework for climate warming projections.
- Engineered models in Fortran and analyzed large NetCDF datasets using Python in Linux.
- Statistical analysis and visualization of ecological time-series data using R.

Job Search

Kerala, India

04/2021 - 07/2021 Kerala, India

Relocation from India to Germany and active job search.

Physics Educator

Tutorwaves Solutions Inc / Freelance

10/2018 - 03/2021 Kerala, India

Prepared physics content for competitive examinations and coached state-level chess teams.

Ph.D. Researcher in Astrochemistry

University of Paris-Saclay

10/2015 - 09/2018 Orsay, France

Led atomic collision experiments, analyzed data, and modeled outcomes.

- Acquired and processed terabytes of raw data from tandem particle accelerator experiments.
- C++ signal processing, algorithm development, and molecular fragmentation modeling (KIDA database).

EDUCATION

Ph.D. in Astrochemistry

Université Paris-Saclay, Institute of Molecular Sciences (ISMO) Orsay, France

09/2015 - 09/2018 Orsay, France

M.Sc. in Physics

National Institute of Technology Calicut, India

07/2012 - 01/2015 Calicut, India

B.Sc. in Physics

University of Calicut, India

06/2009 - 04/2012 Thrissur, India

REFERENCES

Dr. Andreas Busjahn
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Prof. Dr. Kai Wirtz
kai.wirtz@hereon.de

Prof. Dr. Elisa Schaum
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and molecular biology.

OTHER SELECTED PROJECTS

Hepatitis Delta Virus Pipeline • GitHub

Nextflow (DSL2) pipeline for viral sequence analysis: NCBI Entrez download, MAFFT alignment, trimAl trimming, Conda-managed dependencies.

Introduction to Tidymodels • GitHub

Teaching script for ML in R: recipes, parsnip, workflows, and yardstick on the Palmer Penguins dataset.

Berlin Tree Analysis • GitHub

Interactive Jupyter notebook analyzing 5 years of tree planting data in Berlin using Python, Pandas, and Seaborn.

PUBLICATIONS

5 peer-reviewed articles • Google Scholar Profile

LANGUAGES

English

Proficient

German

B1 - Goethe Certified

French

Intermediate

Malayalam

Native

Tamil

Proficient

Hindi

Intermediate